

HW1 - STAT 709

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Problem 1. Ex. 1.7.

Let $\nu_i, i = 1, 2, \dots$, be measures on $(\Omega, \mathcal{F})$ and $a_i, i = 1, 2, \dots$, be positive numbers. Show that $a_1\nu_1 + a_2\nu_2 + \dots$ is a measure on $(\Omega, \mathcal{F})$.

Sol.

Denote $\sum_{i=1}^{\infty} a_i\nu_i =: \nu$ and to prove $\nu(\emptyset) = 0$, we have

$$\nu(\emptyset) = \sum_{i=1}^{\infty} a_i\nu_i(\emptyset) = 0$$

and for any disjoint $\{E_i\}_{i=1}^{\infty} \subset \mathcal{F}$, we have

$$\begin{aligned} \nu\left(\bigsqcup_{i=1}^{\infty} E_i\right) &= \sum_{j=1}^{\infty} a_j\nu_j\left(\bigsqcup_{i=1}^{\infty} E_i\right) \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_j\nu_j(E_i) \\ &= \sum_{i=1}^{\infty} \nu(E_i) \end{aligned}$$

by exchanging the summation, which can be shown by taking the partial summations $\sum_{i=1}^n \sum_{j=1}^n a_j\nu_j(E_i)$.

Problem 2. Ex. 1.8.

Let $\{A_n\}$ be a sequence of events on a probability space $(\Omega, \mathcal{F}, P)$.

Define $\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i$ and $\liminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i$.

Show that $P(\liminf_n A_n) \leq \liminf_n P(A_n)$ and $\limsup_n P(A_n) \leq P(\limsup_n A_n)$.

Sol.

Notice

$$\bigcap_{i=n}^{\infty} A_i \text{ increasing, and } \bigcup_{i=n}^{\infty} A_i \text{ decreasing,}$$

we have

$$P\left(\liminf_n A_n\right) = \limsup_n P\left(\bigcap_{i=n}^{\infty} A_i\right) \leq \limsup_n \inf_{k \geq n} P(A_k) = \liminf_n P(A_n)$$

and

$$P\left(\limsup_n A_n\right) = \liminf_n P\left(\bigcup_{i=n}^{\infty} A_i\right) \geq \liminf_n \sup_{k \geq n} P(A_k) = \limsup_n P(A_n).$$

Problem 3. Ex. 1.10.

Let $F(x_1, \dots, x_k)$ be a c.d.f. on $\mathbb{R}^k$. Show that

- (a) $F(x_1, \dots, x_{k-1}, x_k) \leq F(x_1, \dots, x_{k-1}, x'_k)$ if $x_k \leq x'_k$.
- (b) $\lim_{x_i \rightarrow -\infty} F(x_1, \dots, x_k) = 0$ for any $1 \leq i \leq k$.
- (c) $F(x_1, \dots, x_{k-1}, \infty) = \lim_{x_k \rightarrow \infty} F(x_1, \dots, x_{k-1}, x_k)$ is a c.d.f. on $\mathbb{R}^{k-1}$.

Sol.

There is a probability measure P on $\mathbb{R}^k$ such that

$$F(x_1, \dots, x_{k-1}, x_k) = P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x_k])$$

(a) We know

$$\begin{aligned} F(x_1, \dots, x_{k-1}, x_k) &= P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x_k]) \\ &\leq P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x'_k]) \\ &= F(x_1, \dots, x_{k-1}, x'_k). \end{aligned}$$

(b)

$$\begin{aligned} \lim_{x_i \rightarrow -\infty} F(x_1, \dots, x_k) &= \lim_{x_i \rightarrow -\infty} P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x_k]) \\ &= P\left(\bigcap_{x_i \rightarrow -\infty} (-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x_k]\right) \\ &= 0. \end{aligned}$$

(c) Define $\bar{P}$ to be the probability measure on $\mathbb{R}^{k-1}$ such that

$$\bar{P}((-\infty, x_1] \times \dots \times (-\infty, x_{k-2}] \times (-\infty, x_{k-1}]) = P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times \mathbb{R})$$

on $\mathbb{R}^{k-1}$, and whose existence can be secured by Carathéodory's Theorem. Then

$$\begin{aligned} F(x_1, \dots, x_{k-1}, \infty) &= \lim_{x_k \rightarrow \infty} P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times (-\infty, x_k]) \\ &= P\left((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times \bigcup_{x_k \rightarrow \infty} (-\infty, x_k]\right) \\ &= P((-\infty, x_1] \times \dots \times (-\infty, x_{k-1}] \times \mathbb{R}) \\ &= \bar{P}((-\infty, x_1] \times \dots \times (-\infty, x_{k-2}] \times (-\infty, x_{k-1}]) \end{aligned}$$

and we are done.

Problem 4.

For a matrix $A \in \mathbb{R}^{n \times m}$, recall that the operator norm is the largest singular value: $\|A\|_{\text{op}} = s_{\max}(A)$. Another useful norm is the Frobenius norm, defined by

$$\|A\|_F = \left(\sum_{i=1}^n \sum_{j=1}^m A_{ij}^2 \right)^{1/2}.$$

Let $r = \text{rank}(A)$, and hence $1 \leq r \leq (n \wedge m)$. Write the nonzero singular values as $s_1(A) \geq \dots \geq s_r(A) > 0$.

(i) Prove that $\|A\|_F^2 = \sum_{i=1}^r s_k(A)^2$.

(ii) Prove that $\|A\|_{\text{op}} \leq \|A\|_F \leq \sqrt{r} \|A\|_{\text{op}}$.

(iii) Show that $\|A\|_{\text{op}} = \|A\|_F$ if and only if $r = 1$. Characterize the nonzero eigenvalues of AA^T and $A^T A$ when $\|A\|_F = \sqrt{r} \|A\|_{\text{op}}$.

Sol.

(i) Consider the SVD of $A = QSV$ where $Q \in \mathbb{R}^{n \times n}$, $V \in \mathbb{R}^{m \times m}$ are orthogonal matrices. Then

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m A_{ij}^2 &= \sum_{i=1}^n |A_i|^2 \\ &= \sum_{i=1}^n |(Q_i S V)|^2 \\ &= \sum_{i=1}^n \sum_{k=1}^n q_{ik}^2 s_k^2 = \sum_{k=1}^n s_k^2 = \sum_{k=1}^r s_k(A)^2 \end{aligned}$$

since Q, V orthogonal and denote $s_k := s_k(A)$ for $1 \leq k \leq r$ and $s_k := 0$ for $r < k \leq n$.

(ii) Notice

$$\|A\|_{\text{op}}^2 = s_{\max}^2(A) \leq \sum_{i=1}^r s_i^2(A) = \|A\|_F^2 \leq r s_{\max}^2(A) = r \|A\|_{\text{op}}^2$$

and we are done.

(iii) If $r = 1$, then obviously

$$\|A\|_{\text{op}}^2 = s_1(A)^2 = \|A\|_F^2.$$

For the other direction, we know in this case we have

$$\|A\|_{\text{op}}^2 = s_{\max}^2(A) = \sum_{i=1}^r s_k^2(A) = \|A\|_F^2$$

that means $s_k(A) = 0$ for all $k \geq 2$ if $r \geq 2$, which is a contradiction and hence $r = 1$.

If $\|A\|_F = \sqrt{r} \|A\|_{\text{op}}$, we know

$$r s_1(A)^2 = \sum_{i=1}^r s_i^2(A)$$

and hence $s_1(A) = s_2(A) = \dots = s_r(A) = \|A\|_{\text{op}}$ and since

$$AA^T = QSS^TQ^T, \quad A^T A = V^T S^T S V$$

and we know the nonzero eigenvalues of AA^T and $A^T A$ will all equal to $s_1(A)^2 = \|A\|_{\text{op}}^2$.

Problem 5.

Recall the example from the lecture: let $U \sim \text{Unif}([0, 2\pi])$, and define $X = \cos U$ and $Y = \sin U$. We want to describe their joint distribution $\mu_{X,Y}$ on $\mathbb{R}^2$.

(1) Define $f(u) = (\cos u, \sin u)$ for $u \in [0, 2\pi]$. Prove that f is measurable as a map:

$$f : ([0, 2\pi], \mathcal{B}([0, 2\pi])) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)).$$

You may cite an appropriate theorem. Use the definition of an induced measure to express $\mu_{X,Y}(B)$ for $B \in \mathcal{B}(\mathbb{R}^2)$.

(2) Alternatively, construct the distribution as a measure on the unit circle. Let

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}, \quad A_{u,v} = \{(\cos z, \sin z) : z \in (u, v)\}, u < v.$$

Let $\mathcal{B}(S)$ be the σ -algebra generated by all sets $A_{u,v}$, where $u, v \in \mathbb{R}$ and $u < v$.

(i) Prove that there is a unique probability measure μ_0 on $(S, \mathcal{B}(S))$ such that

$$\mu_0(A_{u,v}) = \frac{v - u}{2\pi} \quad \text{for every } 0 < u < v < 2\pi.$$

You may cite an appropriate measure extension or uniqueness theorem.

(ii) For every $B \in \mathcal{B}(\mathbb{R}^2)$, show that $B \cap S \in \mathcal{B}(S)$. Then show that

$$\mu(B) := \mu_0(B \cap S), B \in \mathcal{B}(\mathbb{R}^2),$$

defines a probability measure on $\mathbb{R}^2$.

(iii) Show that $\mu = \mu_{X,Y}$, where $\mu_{X,Y}$ is the measure from part (1).

Sol.

(1) Since f is continuous, so f is measurable relative to Borel σ -algebra. The conclusion maybe found on any measure theory textbook so I will skip that.

Notice

$$\mu_{X,Y}(B) = P((X, Y) \in B) = P(f(U) \in B) = P(f^{-1}(B))$$

where P is the probability measure corresponding to $\text{Unif}([0, 2\pi])$.

2. (i) For any open interval $(a, b) \subset [0, 2\pi]$, $x \in [0, 2\pi]$, define

$$\rho((a, b)) = \frac{b - a}{2\pi}, \quad \rho(\{x\}) = 0$$

and define

$$\mu^*(B) = \inf \left\{ \sum_{i=1}^{\infty} \rho(I_i), \text{ where } I_i \text{ are open intervals or singleton in } [0, 2\pi], B \subset \bigcup_{i=1}^{\infty} I_i \right\}$$

which is an outer measure on $[0, 2\pi]$, and it is easy to check for any $E \subset [0, 2\pi]$ and I open interval in $[0, 2\pi]$,

$$\mu^*(E) = \mu^*(E \cap I) + \mu^*(E \cap I^c)$$

To check this, we only need to prove LHS greater than RHS, consider $\{E_i\}_{i=1}^{\infty}$ the open intervals cover of E , then we know $E_i \cap I$ is an open interval and $E_i \cap (\bar{I})^c$ are disjoint union of open intervals, with $\mu^*(E_i \cap I) + \mu^*(E_i \cap (\bar{I})^c) = \mu^*(E_i)$ and it will induced the required conclusion. So any open interval are μ^* -measurable, which means $\mathcal{B}([0, 2\pi])$ are μ^* -measurable and μ^* is a measure on $[0, 2\pi]$ by Caratheodory's Theorem.

Now consider the induced measure μ_0 on S , we define $\mu_0(E) := \mu^*(f^{-1}(E))$ for $E \in \mathcal{B}(S)$ where we may prove $f^{-1}(\mathcal{B}(E)) \subset \mathcal{B}([0, 2\pi])$ by π - λ theorem, that is we know the unions of open arcs on S form a π -system and the set E with $f^{-1}(E) \in \mathcal{B}([0, 2\pi])$ will form a λ -system, so μ_0 is well defined on $\mathcal{B}(S)$. Also we have

$$\mu_0(A_{u,v}) = \mu^*(f^{-1}(A_{u,v})) = \mu^*((u, v)) = \frac{v - u}{2\pi}.$$

If there is another measure ν satisfies the equality, then we use the π - λ Theorem again by considering

$$\mathcal{M} = \{E \in \mathcal{B}(S), \mu_0(E) = \nu(E)\}$$

and we know the unions of closed arcs form a subset of $\mathcal{M}$ and a π -system at the same time. We would like to show $\mathcal{M}$ to be a λ -system, $S \in \mathcal{M}$ and if $A \subset B$, $A, B \in \mathcal{M}$, then $\mu_0(B - A) = \mu_0(B) - \mu_0(A) = \nu(B) - \nu(A) = \nu(B - A)$ obviously. For $A_i \uparrow A$, then we know

$$\mu_0(A) = \limsup \mu_0(A_i) = \limsup \nu(A_i) = \nu(A)$$

and hence $\mathcal{M}$ is a λ -system. Therefore, $\mu_0 = \nu$ for any set in $\mathcal{B}(S)$ are we are done.

(ii) We use the notation $\mathcal{A}$ to denote the intersections of some closed balls in $\mathbb{R}^2$ and we know $B \cap S$ is a closed arc and hence in $\mathcal{B}(S)$ for any closed ball B . Now consider $\mathcal{P} = \{B \in \mathcal{B}(\mathbb{R}^2), B \cap S \in \mathcal{B}(S)\}$ and we know $\mathcal{A} \subset \mathcal{P}$ with the frontier a π -system, so we prove $\mathcal{P}$ to be a λ system, and obviously $\mathbb{R}^2 \in \mathcal{P}$. For $A \subset B$, $A, B \in \mathcal{B}(\mathbb{R}^2)$, we know $(B - A) \cap S = B \cap S - A \cap S$ and for $A_i \uparrow A$, $A \cap S = \bigcup_{i=1}^{\infty} (A_i \cap S)$ and hence $\mathcal{P}$ is a λ -system. Therefore, $\mathcal{B}(\mathbb{R}^2) = \sigma(\mathcal{A}) \subset \mathcal{P}$ and we are done.

(iii) For any $B \in \mathcal{B}(\mathbb{R}^2)$, we know

$$\mu(B) = \mu_0(B \cap S) = \mu^*(f^{-1}(B \cap S)) = P(f^{-1}(B)) = \mu_{X,Y}(B)$$

and we are done,, where $\mu^* = P$ on $\mathcal{B}([0, 2\pi])$ with π - λ techniques and we skipped that proof.